SDLC Model Comparison Matrix

Queueless – Smart Queue & Appointment Management System

Comparison based on the criteria of SDLC. It instructs comparing SDLC models according to project characteristics such as project size, requirement clarity, risk level, time constraints, budget, customer involvement, and team expertise.

Selection Criteria	Waterfall	V-Model	Iterative	Incremental	Spiral	Agile
Suitable for changing requirements	No	No	Yes	Yes	Yes	Yes
Suitable for rapid delivery	No	No	Yes	Yes	No	Yes
Supports continuous user feedback	No	No	Yes	Yes	Yes	Yes
Suitable for small/medium academic project	Yes	Yes	Yes	Yes	No	Yes
Requires high-risk analysis	No	No	No	No	Yes	No
Suitable when requirements are well-defined	Yes	Yes	Yes	Yes	Yes	Yes
Supports incremental development	No	No	Yes	Yes	Yes	Yes
Suitable for limited project time	No	No	Yes	Yes	No	Yes
Strong emphasis on testing	No	Yes	Yes	Yes	Yes	Yes
Requires high customer involvement	No	No	Yes	Yes	Yes	Yes

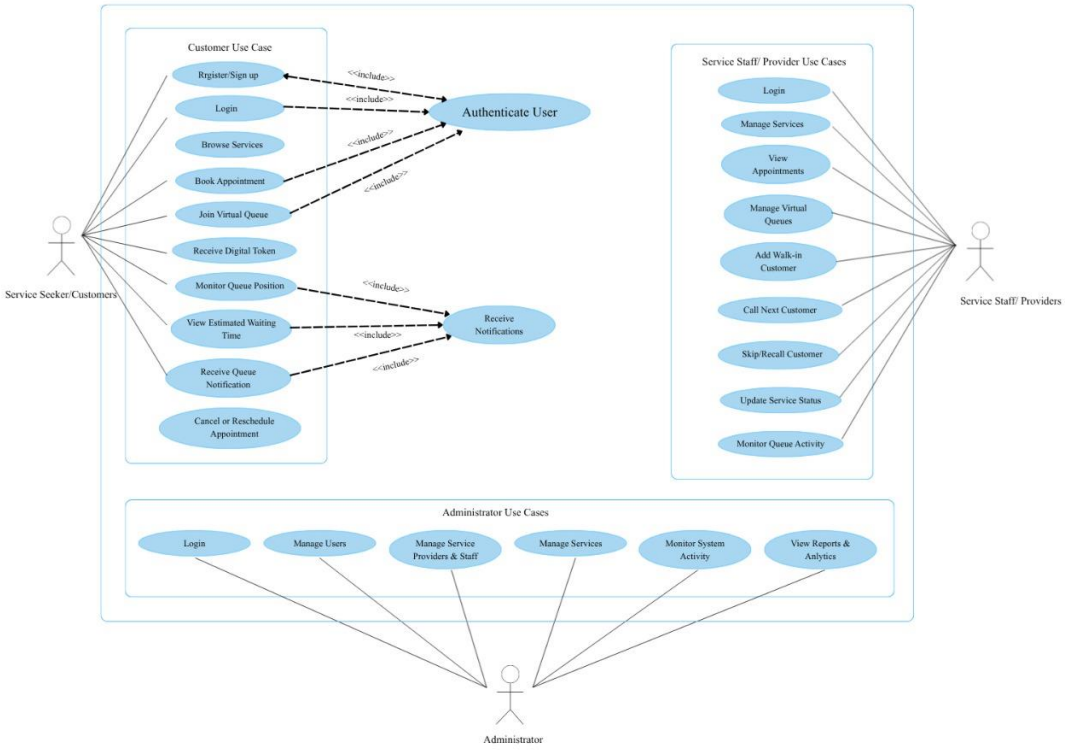
Selected Model: Agile

Agile is selected for Queueless because it is flexible and adaptive, supports continuous feedback and rapid delivery, and is suitable for a project where features can be developed incrementally within an academic time constraint.

Discussion: Queueless is a web-based Smart Queue and Appointment Management System. Its initial version focuses on core queue and appointment functionality, while advanced features are excluded. The project has three main user roles—Service Seeker/Customer, Service Staff/Provider, and Administrator. Therefore, an Agile approach is appropriate because the system can be developed in smaller increments and improved through feedback during development.

Final Justification: Agile is the most suitable SDLC model for Queueless because it provides flexibility, supports changing requirements, enables rapid delivery, and allows continuous feedback and testing.

Design of a Use Case Diagram for a Software System Defining User Stories



The Use Case Diagram of the Queueless – Smart Queue & Appointment Management System

represents how different users interact with the system. The diagram has three main actors: Service Seeker/Customer, Service Staff/Provider, and Administrator.

1. Service Seeker / Customer

The Service Seeker/Customer uses the system to obtain services without having to wait physically in a long queue.

The customer can:

- Register/Sign Up – create a new account.
- Login – access the system using their account.
- Browse Services – view the available services.
- Book Appointment – schedule an appointment for a service.
- Join Virtual Queue – enter a queue digitally instead of waiting physically.
- Receive Digital Token – receive a token number after joining the virtual queue.

- Monitor Queue Position – check their current position in the queue.
- View Estimated Waiting Time – see the expected waiting time.
- Receive Queue Notification – receive notifications about queue progress.
- Cancel or Reschedule Appointment – cancel an appointment or change its scheduled time.

The diagram shows «include» relationships between some customer activities and Authenticate User, meaning authentication is included when performing those activities. Queue-related activities are also connected to Receive Notifications.

2. Service Staff / Provider

The Service Staff/Provider manages the actual service process and customer queue.

The staff can:

- Login – access the staff side of the system.
- Manage Services – manage the services they provide.
- View Appointments – see scheduled customer appointments.
- Manage Virtual Queues – manage customers waiting in the virtual queue.
- Add Walk-in Customer – add customers who arrive without an online appointment.
- Call Next Customer – call the next customer in the queue.
- Skip/Recall Customer – skip a customer or recall them when necessary.
- Update Service Status – update the current service status.
- Monitor Queue Activity – monitor the overall queue and customer flow.

The solid lines connecting the staff actor to these use cases show that these functions are available to the Service Staff/Provider.

3. Administrator

The Administrator manages the overall Queueless system.

The administrator can:

- Login
- Manage Users
- Manage Service Providers & Staff
- Manage Services
- Monitor System Activity
- View Reports & Analytics

These functions allow the administrator to control users, staff/providers, services, and overall system information.

4. System Boundary

The large outer rectangle represents the Queueless system. The use cases are placed inside the system boundary because they are functionalities provided by the system, while the actors are outside because they interact with the system.

5. Relationships

The diagram mainly uses two types of relationships:

- Association (solid line): shows that an actor interacts with a particular use case.
- «include» (dashed arrow): shows that one use case includes another use case as part of its process. For example, Join Virtual Queue includes Authenticate User.